

Final Individual Project: Screen Time and BMI Percentile

Simple Linear Regression using YRBS 2007



Dataset
YRBS 2007



Sample size
n = 12,844



Significance level
 $\alpha = 0.05$

1 RESEARCH QUESTION



Is screen time associated with BMI percentile among students?

2 DATA & VARIABLES



Dataset: YRBS 2007



Sample size: n = 12,844



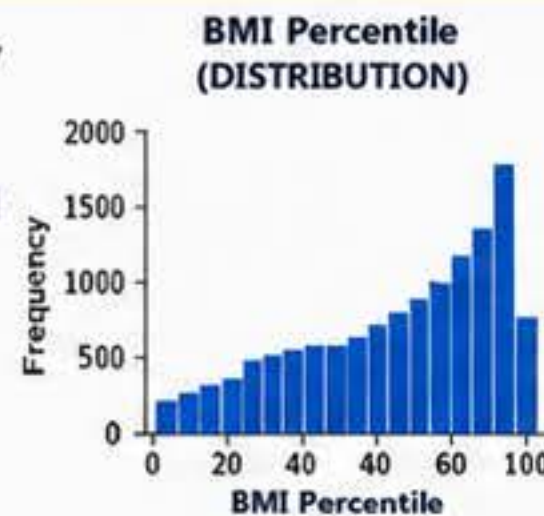
Explanatory variable (X):
Screen time hours



Response variable (Y):
BMI percentile

3 EDA HIGHLIGHTS

- ✓ BMI percentile spans nearly the full 0–100 range.
- ✓ Screen time is concentrated at lower to moderate hour values.
- ✓ The raw scatter suggests only a weak upward trend.



4 METHOD



Simple linear regression

Model: $\text{BMI percentile} = \beta_0 + \beta_1 (\text{screen time hours}) + \epsilon$

Goal: estimate whether β_1 is different from 0.

7 INTERPRETATION / TAKEAWAY

- ✓ Each additional hour of screen time is associated with about 0.84 points higher BMI percentile on average.
- ✓ The association is statistically significant ($p < 0.001$).
- ✓ However, the effect is weak in practical terms.
- ✓ R-squared = 0.0057 means screen time explains only 0.57% of the variation in BMI percentile.

8 DIAGNOSTICS / LIMITATIONS

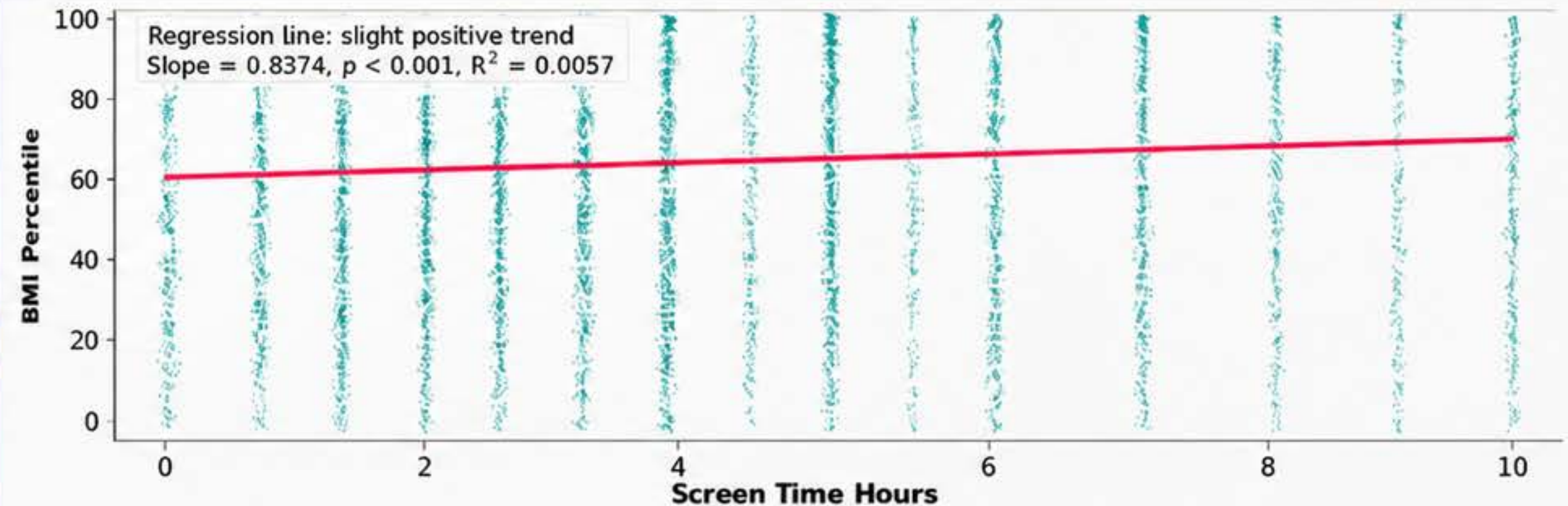
- ⚠ Residual diagnostics suggest deviations from perfect normality.
- ⚠ BMI percentile and screen time are both discrete / limited-format measures.
- ⚠ Screen time is self-reported.
- ⚠ Observational data show **association, not causation**.

9 CONCLUSION



- ✓ There is a positive association between screen time and BMI percentile.
- ✓ The slope is statistically significant, but the explanatory power is very low.
- ✓ Screen time alone is not a strong predictor of BMI percentile.

6 MAIN ANALYSIS: SCREEN TIME HOURS vs BMI PERCENTILE



Each point represents one student. The regression line shows a slight positive association.

5 MAIN RESULTS

Parameter / Metric	Estimate	t Statistic (for slope)	p-value (for slope)	95% CI (for slope)	R-squared
Intercept (β_0)	61.6837	—	—	—	—
Slope for screen time hours (β_1)	0.8374	8.6015	< 0.001	[0.6466, 1.0282]	0.0057



Fitted line equation (rounded):

Predicted BMI percentile = 61.68 + 0.84 × screen time hours



Takeaway: The results are statistically significant but **practically** weak positive association.

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